OVERLAPPING COMMUNITIES IN NETWORK UNVEIL URBAN COMMUTING STRUCTURE FROM MOBILE PHONE DATA

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ABSTRACT

The relationship between occupation and residence in cities not only affects the scale and distribution of commuting traffic, but also relates to the livability of residents. Mobile phone data allow us to capture the job job-housing relationship and explore the urban commuting structure. This paper proposes a mobile-phone-based methodology to study the commuting pattern of residents and unveil the commuting structure of Shanghai. After identifying the residents and commuters and extract their home and job activity points, we built a network on traffic analysis zones based on their job-housing relationships and apply overlapping community detection method to divided the area of Shanghai into closely related sub-regions. By decomposing job-housing networks, three levels of urban commuting structure are described by three types of communities, including: large communities showing multiple nuclei structure of the city, small communities showing short distance commuting communities and banded communities showing long distance commuting demand. This study provides insights into commuting patterns and city commuting structures, which can help to understand the residential commuting demand and support urban transportation planning.

Key words: urban commuting structure; mobile phone data; job-housing relationship; overlapping community detection; network analysis

1 INTRODUCTION

The relationship between occupation and residence in cities not only affects the scale and distribution of commuting traffic, but also relates to the livability of residents. As we move toward trying to 'design demand' rather than simply trying to serve demand, studying the commuting pattern and depicting the job-housing relationship become crucial. Understanding the logic behind the forming of city commuting structure can help us designing a better city and leading sustainable transportation demand.

The emerging of pervasive, geospatial data generated by individuals have recently triggered an opportunity for studying individual mobility patterns (*1*), city dynamics (*2*) and city structure (*3*). As a new travel survey tool, mobile phone data is more pervasive and accurate, which allow us to fully track the trip chain of individual in both temporal and spatial dimensions. It offers a new way to study the job-housing relationship and urban commuting structure.

In our previous research, we have used the modularity based fast unfolding algorithm to decompose the job-housing relationship network into non-overlapping communities and discovered a two-level hierarchical city structure(*4*). The previous research have proved that non-overlapping community detection allow us to discover the hierarchical commuting structure.

Based on the previous study, the purpose of this study is to further explore urban commuting structure from the job-housing function of the city. We introduce the overlapping community detection to decompose job-housing network and analyze job-housing relationships extracted from mobile phone data of Shanghai, China.

2 LITERATURE REVIEW

There are two main approaches for assessing the structure of city-regions (*5*). One is morphological approach, which employs the attributes or internal characteristics of centers, such as the number of jobs (*6*). The morphological approach assesses the city structure on the spatial pattern, with the balance in the size distribution or distribution of absolute importance of centers based on the data from field surveying, remote sensing and policy consulting.

The other is functional approach, classifying metropolitan spatial structure based on the structure of flows within spatial systems (*5*). The functional approach believes that the underlying structure of a city is determined by the flows of people, freight, money and information, which connect the discrete places into an integrated system. In contrast to the space of geospatial, these flows construct the 'space of flows' (*7; 8*). More and more scholars are trying to capture the structure inside large city or even the interaction between cities by studying flows using new sources of data. The new sources of data include: public transportation card data (*9*), taxi trip data (*3*), business services network data (*10*), etc. However, because of the limitations of data access, analytic tools, and computation capabilities, studies of human travel flows had limited development.

Networks have become a key approach to understanding the job-housing interaction inside the city. In network analysis (*11*), a community is a collection of highly inter-connected nodes. Decomposing the network into different communities can retrieve comprehensive information of the structure in the network. Community detection is a subject aiming to identify communities from networks. There are mainly two types of approaches for community detection. One is non-overlapping community detection, in which each of the nodes in network only belong to one particular community. A huge variety of community detection techniques have been developed, based variously on centrality measures, flow models, random walks, resistor networks, modularity

optimization, and many other approaches(*12-14*). The other type of approaches is overlapping community detection, in which believe that Communities in networks often overlap and nodes can simultaneously belong to several communities(*15*). These approaches include clique percolation method(*16*), local optimization of fitness function(*17*), and clustering link communities(*15*).

As a new travel survey tool, mobile phone data has attracted the attention of more and more scholars. They used such data to study human mobility under various situations, which include: real-time movement of population in large events (*18*), Origin-Destination (OD) matrix estimation (*19*), travel chain identification (*20*) and home/workplace inference (*21*), etc. mobile phone data have also been successfully used to study the general law of human mobility (*1*). Thus, mobile phone data has a good potential for studying commuting flows and unveiling urban structure in functional approach.

3 DATA AND STUDY AREA

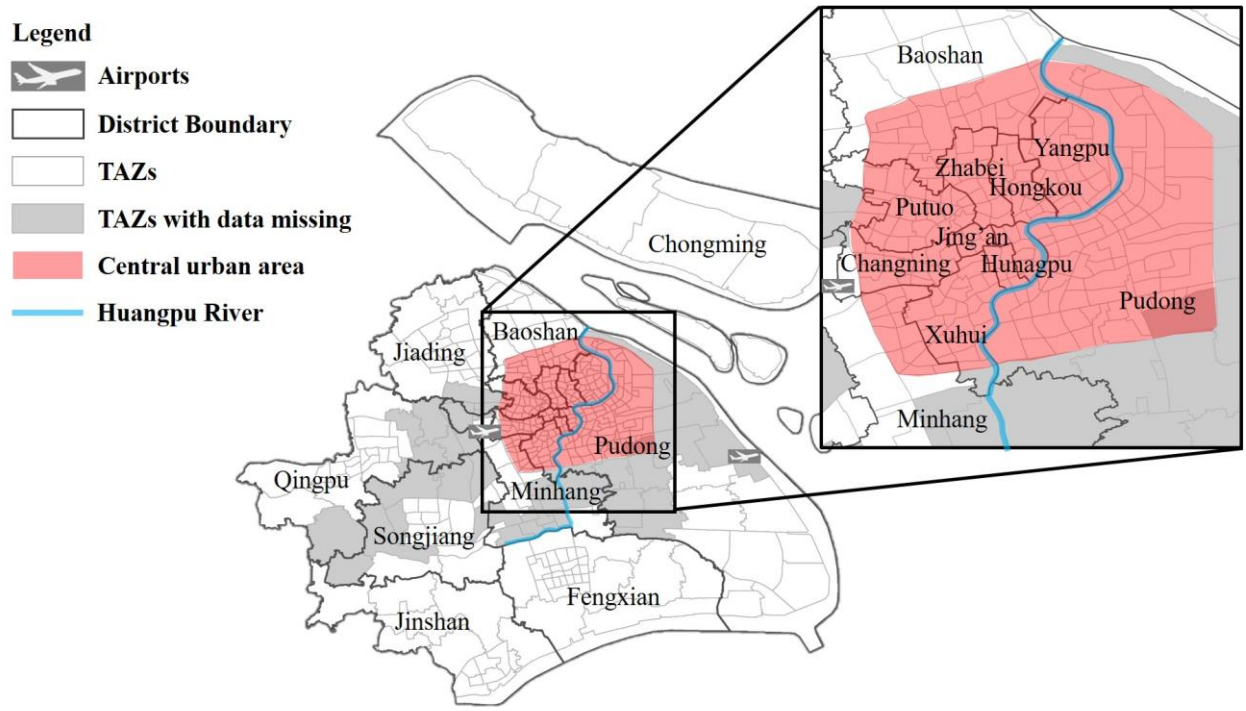
3.1 Mobile phone data

Anonymous mobile phone data used in this paper was collected for billing and operational purposes from September 1 to September 15, 2011 in Shanghai, China. It includes the information of encrypted mobile phone identifier, service time, service type, location of base transceiver station (BTS) and location area (LA), etc. A record of mobile phone data will be generated when a call is placed or received, a text message is sent or received, the phone is switched on or switched off, or the phone signal is handed over from one BTS to the other BTS, etc. The average number of records was 1 billion per day, covering 25 million active users. The coverage radius of BTS is 500 to 800 meters.

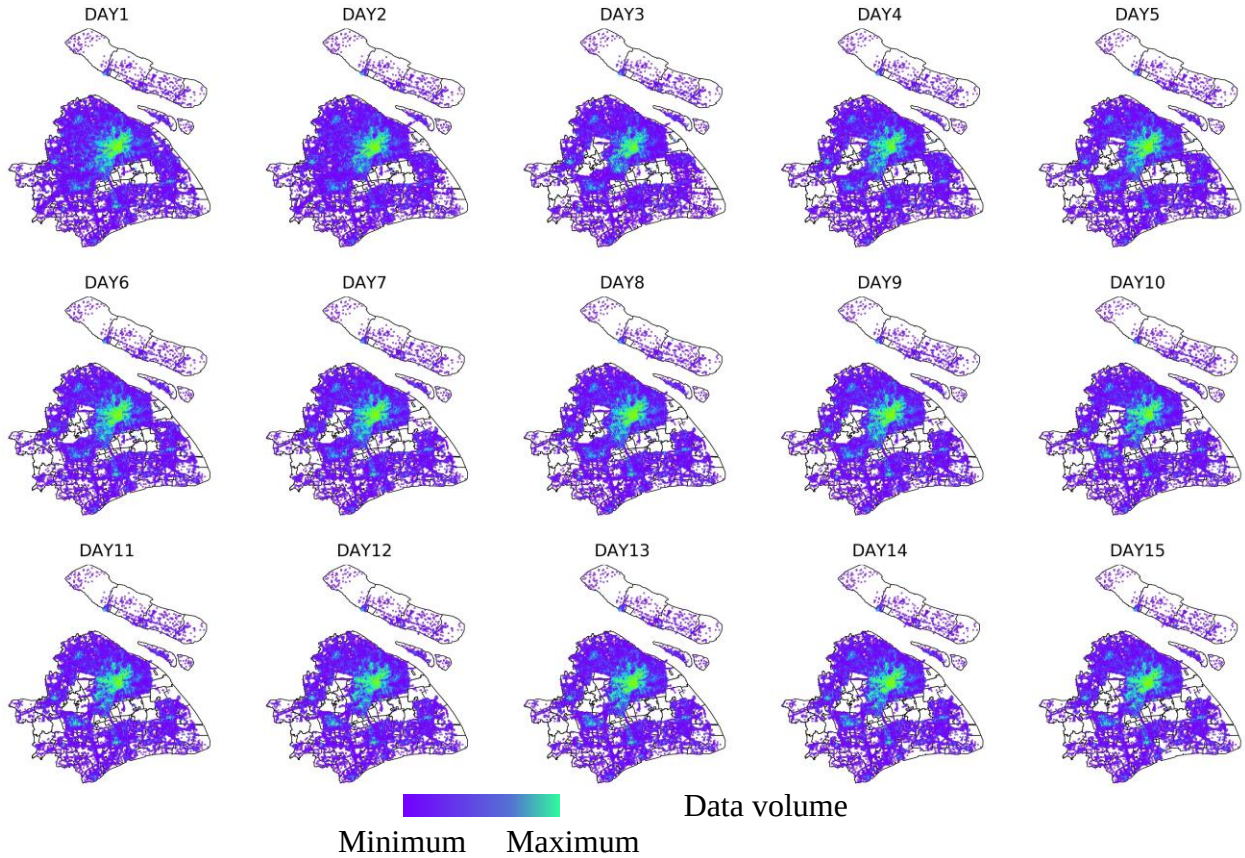
3.2 Study area

Our study area covers all the area of Shanghai. Shanghai is the economic center of China, with the area of 6340 km^2. By the end of 2011, the district of Shanghai was consisted of 16 districts and 1 country (see FIGURE 1(a)). According to the master planning of Shanghai, the central urban area is the area inside the Outer Ring Expressway. The Huangpu River divides Shanghai into two parts: Pudong on the east side and Puxi on the west side. Puxi is the old urban area with a long history, which is also the core urban areas of Shanghai. As a new district, Pudong witnesses a rapid development with the support of policy.

In our study, the area of Shanghai has been divided into 447 traffic analysis zones (TAZs), with the same land use in each zone. However, the data suffering from varying degrees of data missing during the whole observation period(see FIGURE 1(b)). As a result, we have to eliminate 44 TAZs suffering from severe data missing. The remaining 403 TAZs are the main objects of this study.



(a) Study area



(b) Data quality in the observation period

FIGURE 1 Study area and data quality in the observation period

4 METHODOLOGY

4.1 Data preprocessing

Once a user can be captured by more than one BTSs at the same time, its signal will be hand over frequently between these BTSs and generate a great number of records in a very short time even if there is no position changing of the user. The frequent handover leads to data noise and waste of system resources. Therefore, a binning method (*21*) was used to cope with this problem and reduce the volume of data:

1) The area of Shanghai is rasterized into 85,260 (245*348) grids. Each of these grids is 500m*500m square.
2) We calculate the average position of every user in every 10 minutes and define which grid it belongs to. The center of this grid will be regard as the position of the user during the 10-minute period.

After the binning, we can receive a set of control points for each phone user, formulated as:

$$c_i(grid_i, t_i), i = 1,2, \ldots \quad (1)$$

Where c_i refers to the i-th control point of a user; t_i represents the time when the user arrived at the control point.

4.2 Extracting home and job activity points from mobile phone data

4.2.1 Identifying residents and their homes

Due to the large data volume, we choose a simple method proposed by Li et al. (*21*) to identify the home locations of residents from mobile phone data. The method includes two rules:

(a) In the period from 9 p.m. to 9 a.m. the next day, the user stays at one place for no less than six hours.

(b) In our observation periods, the user stays at the place satisfied rule (a) in more than 2/3 of the days (10 days).

If a user satisfied both two rules, the user can be considered as a resident and the place will be considered as the location of home.

4.2.2 Identifying commuters

In previous studies, work places are often considered to be unique for every commuter. The identifying methods of work location always find a fixed place based on the regularity of individual travel pattern during the observation days (*22*). In this way, the commuters who have multiple work locations are neglected.

To avoid this defect, we identify commuters by the following method. After identifying residents, we can obtain their home locations. If a resident appears in a place other than his home, we can regard that he goes out. Commuters should be the residents who stay significantly more time outside home when comparing with non-commuters.

A simple method amounts to choose a threshold δ and to consider that the resident with the average stay time outside home over δ on weekday as a commuter, otherwise, he/her is a

non-commuter. Here we choose the mean of stay time outside home of all residents as the threshold δ, which will split residents into two equal parts — commuters and non-commuters.

4.2.3 Identifying activity points and job activity points

Given the i-th control point and the following k control points where $grid_{i+k} \neq grid_{i+k-1} = \cdots = grid_i \neq grid_{i-1}, k \geq 1$, the activity duration of the user stays at $grid_i$ can be calculated as:

$$d = t_{i+k-1} - t_i \tag{2}$$

According to the household travel survey of Shanghai, 30 minutes can represent the critical station of an individual's daily movement and contribute to the comprehensive understanding of individual activities(*23*). So the location where an individual stays over 30 minutes is defined as an activity point.

However, the mobile phone data do not contain the information of the type of activity.

Activity points at work time (9a.m. – 18p.m. on weekdays) are regard as job activity points.

4.3 4.3 Unveiling city commuting structure using network analysis method

4.3.1 Construction of network

The commute flows within a city connect discrete places into an integrated system. Among TAZs, we can aggregate commute trips to obtain spatial interactions between zones. Constructing a network and apply network analysis methods upon TAZs, we can further understand the urban commute interactions.

For the residents in TAZ i, the number of job activity points they have per hour in TAZ j can be define as the number of connections from TAZ i to TAZ j, which is denoted as Q_{ij} here. By aggregating all job activity points and home location, we can obtain a 403×403 matrix V.:

$$V = \begin{bmatrix} Q_{11} & Q_{12} & \cdots & Q_{1j} \\ Q_{21} & Q_{22} & \cdots & Q_{2j} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{i1} & Q_{i2} & \cdots & Q_{ij} \end{bmatrix} \tag{3}$$

To build a network, each TAZ is represented as a node. Between every node i and j, we constructed two directed edge E_{ij} and E_{ji} with the weight Q_{ij} and Q_{ji}.

4.3.2 Net-inflow index

In order to verify the job activity points we identified, we introduce the net-inflow index to measure whether a TAZ tends to be a job center or a residential community. In the network we constructed, the connection between every two TAZs can be regarded as commuting flow. The commuting flow from TAZ i to TAZ j can be considered as the outflow from TAZ i and the inflow to TAZ j. The net-inflow index for TAZs is defined as:

$$N_i = \sum_i Q_{ji} - \sum_i Q_{ij} \tag{4}$$

As the average value of N_i is 0, a TAZ with $N_i > 0$ means that more job activity points are attracted into it, which means that this TAZ is more tend to be a job center. While a TAZ with $N_i < 0$ means it is more likely a Residential community.

4.3.3 Overlapping community detection

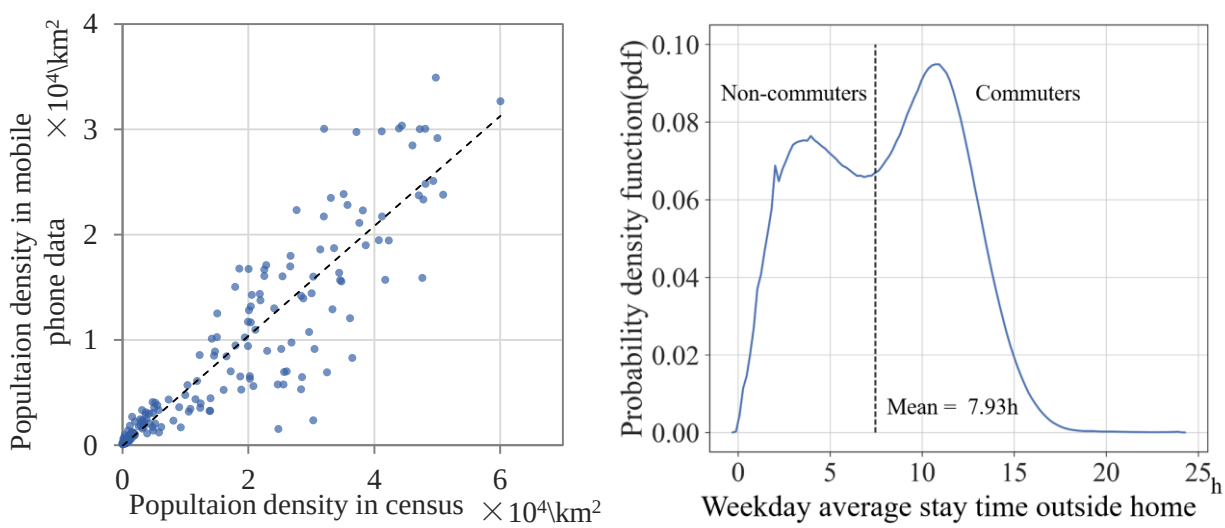
In network analysis (*11*), a community is a collection of highly inter-connected nodes. The nodes belonging to different communities are sparsely connected. In order to retrieve comprehensive information of the structure in the network, we use the overlapping community detection to decompose the housing network into communities, dividing the city into sub-regions with intensely-interactive job-housing relationships.

There are many approaches to discover overlapping communities, here we use the method based on link communities clustering(*15*). The basic concept of this method is assuming that nodes in the network have multiple identities, and they will cluster in corresponding communities according to their identities. In another words, communities are depending on the attribute of the links between its members. Hence, this method clusters the links by measuring the similarity of links. The nodes that connected by the links in the same cluster will be regarded as belonging to the same community. For that there will be several links connecting a single node, with these links being clustered into different clusters, the node can simultaneously belong to several communities.

In this paper, we use the linkcomm package in R to conduct overlapping community detection. This algorithm chooses the Jaccard similarity coefficient to calculate the similarity matrix for links in network, and cluster the links using hierarchical clustering. In order to determine the best cluster number, this algorithm also introduce the index of partition density to measure the connection inside communities(*15; 24*).

5 RESULTS AND DISCUSSION

5.1 Result of residents and commuters identification



(a)Compare of residents identified (b)Distribution of weekday average egression time

FIGURE 2 Result of residents and commuters identification

By the identification methods, we identified 9.86 million residents, accounting for 42% of the total population of over 23.47 million in Shanghai by the end of 2011(*25*). We compare it with the sixth national census in China conducted in 2010. The census was conducted in Jiedao units the smallest administrative divisions in China). Therefore, we compare the population density identified by mobile phone data with the permanent residents in the census in Jiedao level. The correlation coefficient between them is 0.91 (FIGURE 2(a)). Although deviations inevitably exist, the mobile phone data can generally cover residents in the area of Shanghai.

As we mentioned in Section 4.2.2, the choice of threshold δ is significant for commuters identification. From the 9.86 million residents identified, we firstly eliminate the mobile phone users who never move during our observation period (1.13 million users in total). Then, for the remaining 8.73 million residents, we calculate the average stay time outside home on weekday for each users and plot the probability density function in FIGURE 2(b). From FIGURE 2(b), we can see two peaks, one is around 4 hours and another is around 11 hours. 11 hours is a rational stay time outside home for a commuter on working days, i.e., go out at 7 or 8 a.m. and return home at 6 or 7 p.m. The mean of stay time outside home on weekday for all 8.73 million residents is 7.93 hours. Choosing this value as threshold δ can divide the residents into two equal parts – commuters and non-commuters.

5.2 Verifying job activity points identified

We calculate the net-inflow index for all TAZs and identify whether a TAZ is a job center or a residential community. As shown in FIGURE 3, TAZ 1-4 are top 4 central business districts (CBDs) in Shanghai and TAZ 5 and 6 are two bases for high-tech industries; TAZ 7-10 are large residential communities. These results are in accordance with the actual land use. Therefore, the net-inflow index can be used in characterize a TAZ as job center or residential community. And it also verifies the job activity points we identified.

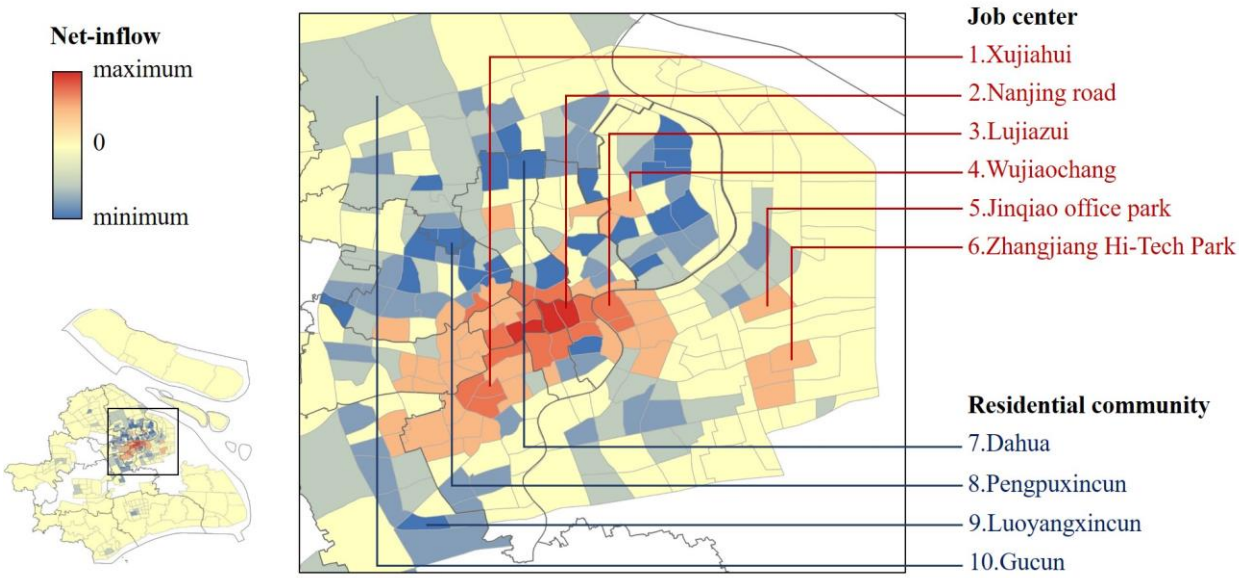


FIGURE 3 Net-inflow index of TAZs

5.3 Overlapping communities

By identifying commuters and extracting their job activity points, we can aggregate them into TAZs and construction the job-housing relationship network. Then we input the network into the link community clustering algorithm. In hierarchical clustering, clusters merge from bottom to top and show a dendrogram of hierarchical structure. With the increase of the height of dendrogram, the clusters are merging into fewer clusters. Finally, all links are merged into a single cluster. In the merging process, the change of partition density is also calculated and shown in.

From the change of the partition density, we can see that there is a gap of the partition density between the height of 0.75 and the height of 0.8. So we choose the result with the height of 0.75 as the result of overlapping community clustering. The result segments the TAZs of Shanghai into 86 communities. Although we have never input any spatial relationship into the algorithm, most of the communities are still merged by adjacent TAZs. The communities provide insights into how the city could be properly divided into closely related sub-regions on the basis of job-housing relationship. Communities in the network represent regions with intense job-housing connection.

Based on the shape of the communities, we classify the 86 communities into three types: large communities(17 communities in FIGURE 5(a)), small communities(66 communities in FIGURE 5(b) and (c)) and banded communities(4 communities in FIGURE 5(d)).

In the large communities in FIGURE 5(a), the first community is constructed by the TAZs in city center. The other 16 communities are all in the suburban area. These large communities show the area of central urban district and the towns in suburban area based on the job-housing relationship. In urban transportation planning, this result can help determining the planning area. In suburban area, the boundary of communities is mostly in correlation with the boundary of administration districts. As comparing to our previous research(*4*), the result is similar. In central urban area, the public transportation systems are well developed, which makes cross-regional employment easy. But in suburban district, due to poor cross-regional traffic connection, cross-regional job opportunities are difficult to reach.

When depicting the area of the central urban area and the towns in suburban area, the overlapping communities also depict the small communities with intense job-housing connection inside the large communities, as is shown in FIGURE 5(b) and (c). Small communities are mostly distributed in urban central areas, which indicates that there are a large number of short distance commuters in the central area of the city. These commuters are commuting within a small area of 3-5 TAZs. We also compare it with the immigration data collected in the sixth national census conducted in 2010, as is shown in FIGURE 6. In recent years, there are less immigrations moving into the central urban area of Shanghai. Most of the residents in the central urban area are natives and they have bought their house before the steep rising of the housing prices. Their living places are close to their working places, with small commute distance and forming the small communities.

At the same time, there are also 4 banded communities in the overlapping communities indicating the long distance commuting demand(FIGURE 5(d)). From FIGURE 6, at the fringe of central urban area, the population is consisted with high proportion of immigrations and forming into large-scale residential communities. The 4 banded communities are shaped as sector structure radiating out from the city center to the residential communities at the fringe of central urban area. Comparing the communities with the metro lines in Shanghai, the banded communities are all extend outward along with radiating metro lines, with averagely two metro lines to form a banded community. From this structure, we can infer that the commuting behavior of residents living along the metro lines depend heavily on the metro line, with their work places aggregate along the metro

line. This result demonstrates that, as the major traffic corridors, metro lines are playing important roles in forming city commuting structure.

In summary, from the result of overlapping communities, we can simplify the commuting structure of Shanghai as follow:

- On the city scale, there is a multiple nuclei structure, with the central urban area as the largest center and several centers of large communities in suburban area.
- Inside these multiple centers, there are many small communities with intense job-housing connection, most of them are in central urban area.
- In the central urban area, there are several sector structures radiating out along the metro lines from city center to the fringe, showing long distance commuting demand.

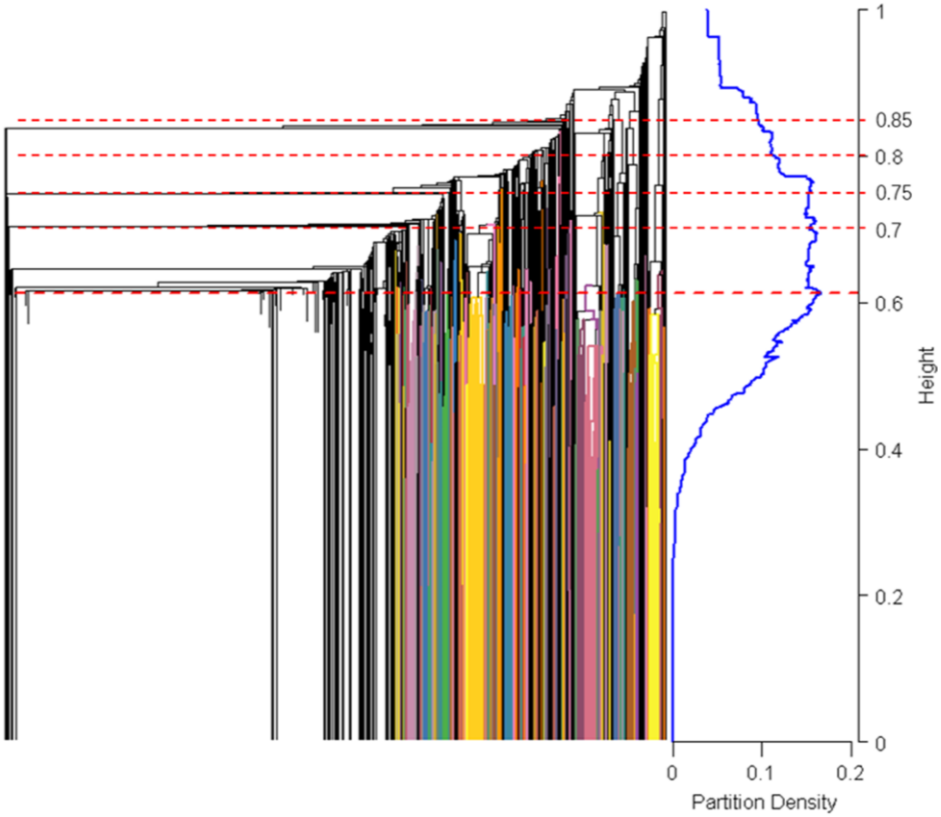
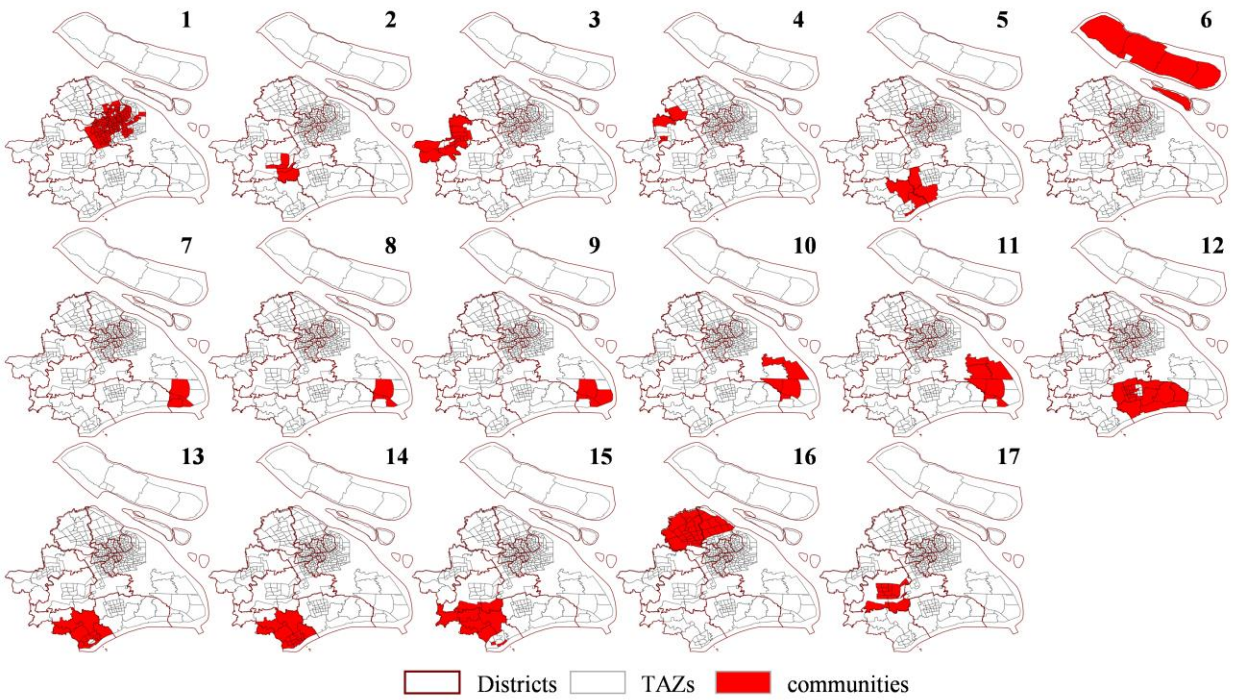
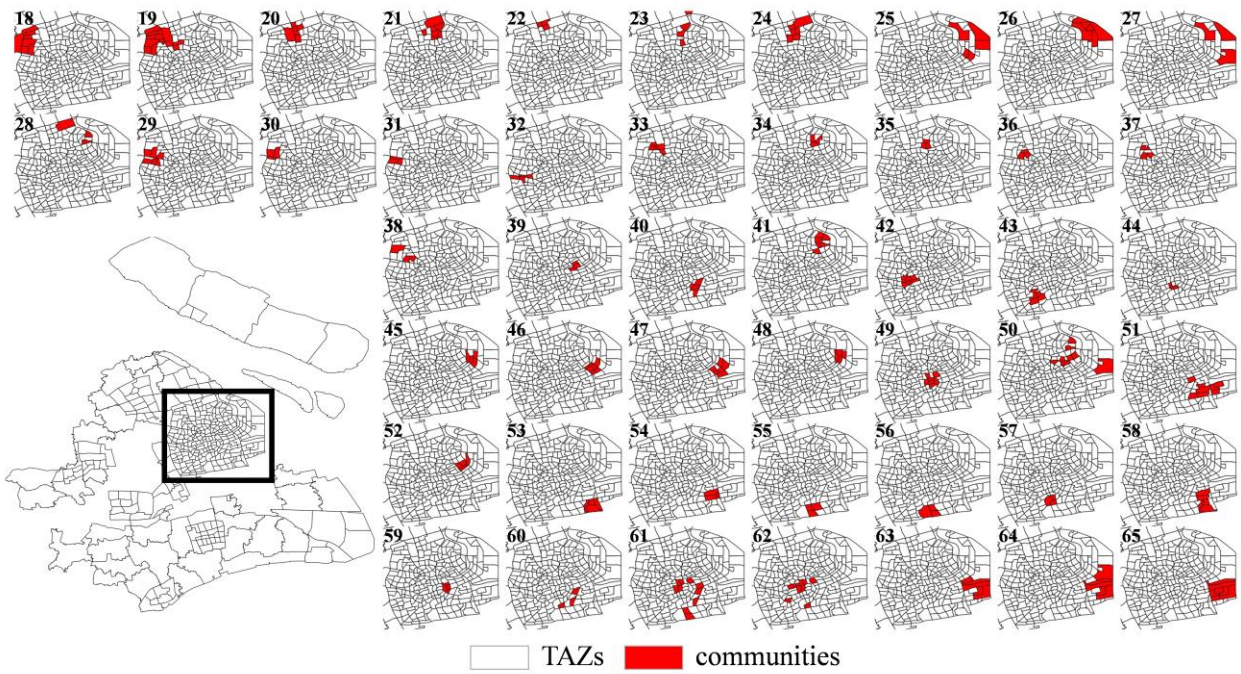


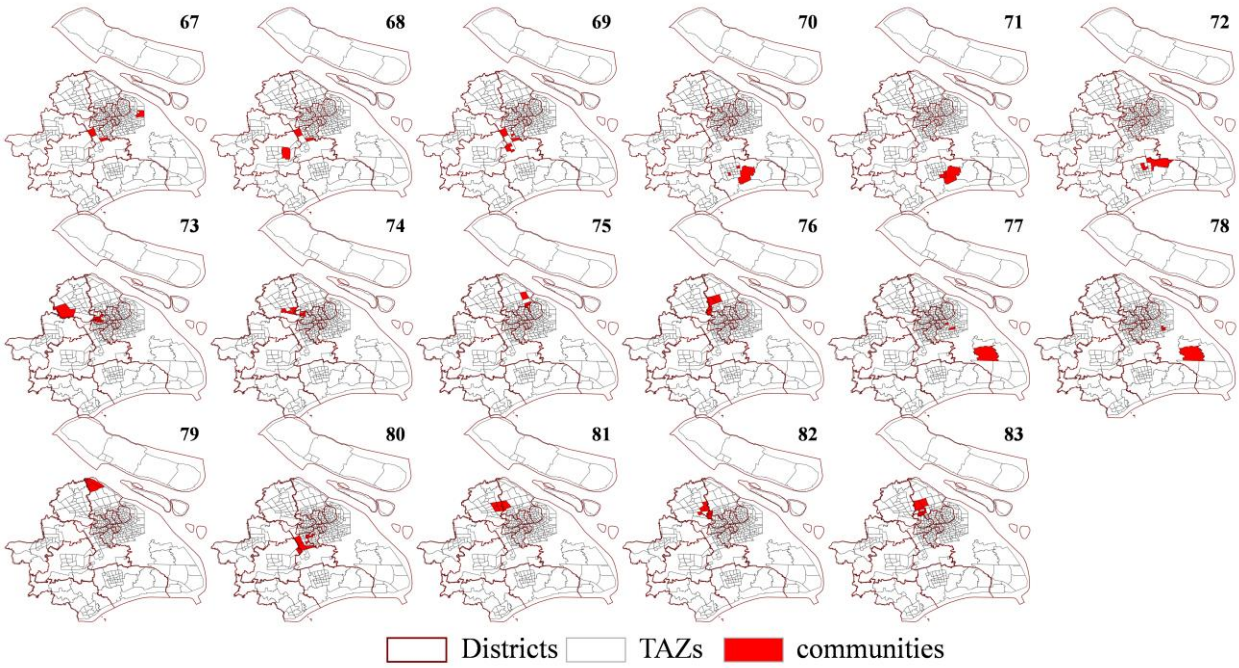
FIGURE 4 Link communities dendrogram of hierarchical clustering



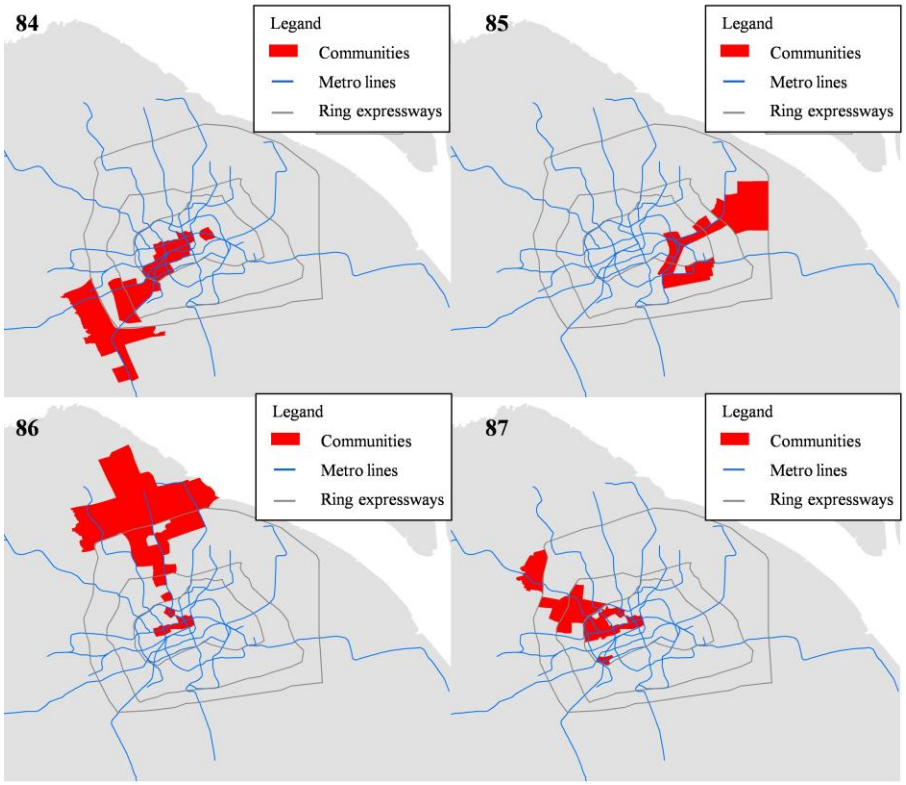
(a) Large communities



(b) Small communities in central urban area



(c) Small communities in suburban area



(d) Banded communities

FIGURE 5 Result of overlapping community detection

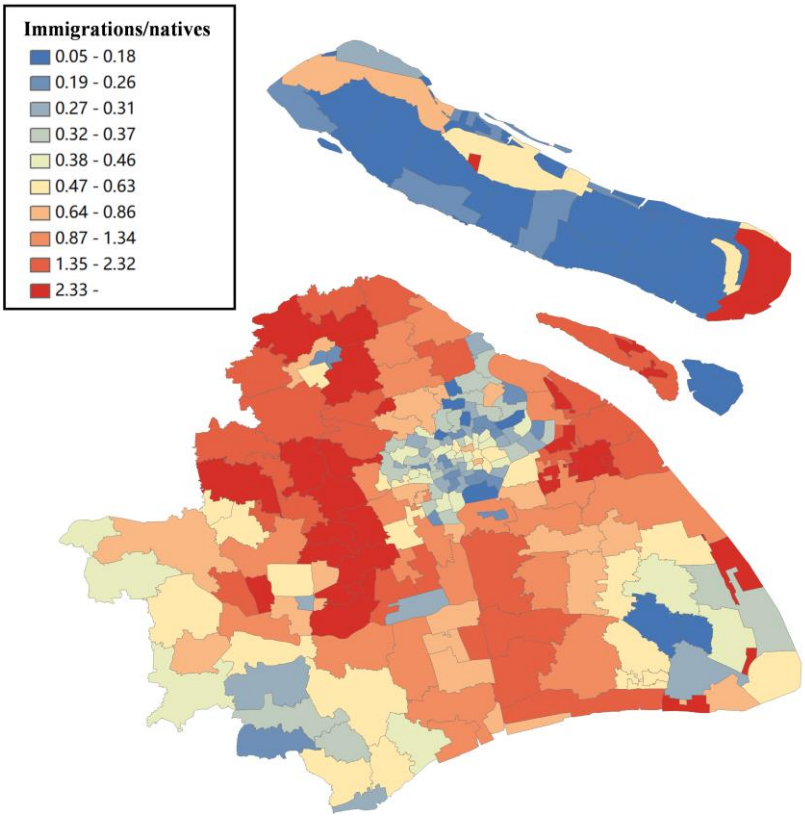


FIGURE 6 Proportion of immigrations to natives from census data

6 CONCLUSION AND CONTRIBUTION

This study proposes methodologies to identify commuters, extract job activity points and unveil city commuting structure from mobile phone data by using overlapping community detection. Taking Shanghai as case study, by identifying overlapping communities, the area of Shanghai is divided into closely related sub-regions based on the job-housing relationship. The conclusion is as follow:

By decomposing job-housing network into three types of communities, different levels of urban commuting structure can be described, including: large communities showing multiple nuclei structure of the city, small communities showing short distance commuting communities and banded communities showing long distance commuting demand.

The contributions of this study include:

1) Proposed a method to extract job anchor points from mobile phone data and proposed the net-inflow index which can be used in identifying job center and residential community;

2) Proposed a method to unveil city commuting structure by constructing complex network and applying overlapping community detection;

3) Describing the commuting structure of Shanghai by divided it into three types of communities that could help city planner further understand the commuting demand in Shanghai.

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AUTHOR CONTRIBUTION

The authors confirm contribution to the paper as follows:
Study conception and design: Qing Yu, Weifeng Li, Dongyuan Yang;
Data collection: Dongyuan Yang;
Analysis and interpretation of results: Qing Yu;
Draft manuscript preparation: Qing Yu.
All authors reviewed the results and approved the final version of the manuscript.

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